Milestone 1 Report

Smart MCQ Solver Challenge

Deep Learning and Generative AI Project

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1. Introduction

The Smart MCQ Solver Challenge focuses on developing intelligent Natural Language Processing (NLP) techniques to automatically identify the most appropriate answer for multiple-choice questions.

The first milestone establishes a classical NLP baseline before introducing advanced transformer-based models in later milestones. The objective is to understand the dataset, preprocess textual information, generate semantic representations, and evaluate baseline retrieval performance.

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2. Objectives

The primary objectives of Milestone 1 are:

- Explore and understand the dataset.
- Analyze the distribution of correct answers.
- Preprocess textual data.
- Generate semantic representations using TF-IDF.
- Compare answer options using cosine similarity.
- Rank answer choices.
- Evaluate predictions using MAP@3.

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3. Dataset Description

The project uses the Smart MCQ Solver Challenge dataset.

Each sample consists of:

- Question prompt
- Five candidate answers (A–E)
- Correct answer (training set)

The testing dataset contains only the prompt and answer options.

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4. Data Exploration

The dataset was initially inspected to understand:

- Number of samples
- Missing values
- Class distribution
- Structure of prompts
- Answer options

Basic exploratory analysis confirmed that the dataset is suitable for semantic similarity based retrieval.

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5. Text Preprocessing

Several preprocessing operations were applied before feature extraction.

These include:

- Lowercasing
- Removing punctuation
- Removing unnecessary whitespace
- Basic text normalization
- Stopword filtering (where applicable)

These preprocessing steps improve the quality of feature extraction while reducing textual noise.

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6. TF-IDF Representation

Term Frequency–Inverse Document Frequency (TF-IDF) was used to convert textual data into numerical feature vectors.

Each question prompt and its candidate answers were transformed into sparse vector representations.

TF-IDF emphasizes words that are important within a document while reducing the influence of frequently occurring terms.

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7. Semantic Similarity

Cosine similarity was computed between the TF-IDF vector of each question prompt and each candidate answer.

The similarity score measures how closely an answer matches the semantic content of the corresponding question.

For every question:

1. Compute similarity with Answer A
2. Compute similarity with Answer B
3. Compute similarity with Answer C
4. Compute similarity with Answer D
5. Compute similarity with Answer E

The answers are then ranked according to their similarity scores.

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8. Baseline Pipeline

The complete baseline pipeline consists of the following stages:

1. Load dataset
2. Preprocess text
3. TF-IDF vectorization
4. Cosine similarity computation
5. Rank answer options
6. Generate Top-3 predictions
7. Evaluate using MAP@3

This pipeline serves as the classical NLP benchmark for future improvements.

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9. Evaluation Metric

Performance is evaluated using Mean Average Precision at 3 (MAP@3).

Rather than considering only the top prediction, MAP@3 rewards models that place the correct answer within the top three ranked options.

This metric is particularly suitable for retrieval-based multiple-choice systems.

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10. Results

The milestone successfully produced:

- Dataset exploration
- Clean preprocessing pipeline
- TF-IDF feature extraction
- Cosine similarity ranking
- Top-3 prediction generation
- MAP@3 evaluation

These results establish a reliable baseline that will be compared against transformer-based approaches in subsequent milestones.

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11. Challenges

Some challenges encountered during implementation include:

- Text normalization across varying question formats.
- Handling sparse TF-IDF representations.
- Efficient computation of similarity scores.
- Improving retrieval quality using only classical NLP methods.

These limitations motivate the transition to contextual embeddings in later milestones.

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12. Future Work

The next milestones will extend this work by introducing:

- Transformer-based language models
- DeBERTa fine-tuning
- Retrieval-Augmented Generation (RAG)
- LoRA-based parameter-efficient training
- Multi-model probability ensembling
- Model deployment

These enhancements aim to significantly improve answer ranking performance beyond the TF-IDF baseline.

13. Conclusion

Milestone 1 successfully established the foundational NLP pipeline for the Smart MCQ Solver project. Through dataset exploration, preprocessing, TF-IDF vectorization, cosine similarity computation, and MAP@3 evaluation, a complete classical baseline has been implemented. This milestone provides the necessary groundwork for the transformer-based architectures and advanced retrieval techniques that will be developed in subsequent milestones.